

Stephan Hartmann: Effective Field Theories, Reductionism and Scientific Explanation

1 Effective Field Theories

Effective theories describe phenomena efficiently, but are incomplete in some way. Effective field theories are usually only applicable in some (low) energy scale. Stephan Hartmann discusses several examples and draws philosophical lessons.

1.1 Examples: Euler-Heisenberg and Fermi theory

Euler-Heisenberg theory: Photon-photon scattering, creating electron-positron pair which decays back into photons (energy-momentum conservation).

Described by quantum electrodynamics (QED): special case of energies well below creation of electron/positron.

Form of effective Lagrangian:

$$\mathcal{L}_{eff} = -\frac{1}{4}F^{\mu\nu}F_{\mu\nu} + \frac{a}{m_e^4}(F^{\mu\nu}F_{\mu\nu}) + \frac{b}{m_e^4}F^{\mu\nu}F_{\nu\sigma}F^{\sigma\rho}F_{\rho\mu} + \mathcal{O}(F^6/m_e^8). \quad (1)$$

How to determine a and b ?

1. From QED directly.
2. Calculate special case exactly and by $\mathcal{L}_{eff}$, compare results.

Euler chose 2, which results in:

$$L_{eff}^{HE} = \frac{1}{2} (E^2 - B^2) + 2 \frac{\alpha_0^2}{45 m_e^4} \left[(E^2 - B^2)^2 + 7 (E \cdot B)^2 \right]. \quad (2)$$

This Lagrangian is much easier to handle than the Lagrangian of the underlying theory, $\mathcal{L}_{QED}$.

Fermi-Theory: Description of the process $n \rightarrow p + e^- + \bar{\nu}_e$.

Given some assumptions, Fermi derived many accurate results. Both his theory and the $V - A$ theory (that was developed on its basis) are incredibly useful, but not renormalisable.

Both theories:

1. Only take relevant fields into account.
2. Are valid only at a given energy scale.
3. Are non-renormalisable.
4. Are based on certain symmetries.
5. Produce scientific understanding.

Both are clearly cases of successful scientific theorizing / scientific progress. What are the philosophical implications of this? How should we think about the relationship between theories, models, and EFTs? How should we view renormalisation?

1.2 Renormalisation

Example QED: Scattering of an electron by an external potential A_{ext} leads to divergent integral for T -matrix:

$$T^{(a)} = -e_0^3 \int_0^\infty d^4k \dots \rightarrow \infty \quad (3)$$

Initial assessment: The theory is inconsistent, because it gives a divergent result.

However:

- Theory was empirically successful.
- Divergences can be avoided by renormalisation scheme.

Renormalisation:

- Redefine parameters $m_0 \rightarrow m_0 + \delta m_0$, $e_0 \rightarrow e_0 + \delta e$.
- Instead of $0 < k < \infty$, consider $\Lambda < k < \Lambda_0$.
- Redefine $e_0 \rightarrow e_o [1 - c_0(\Lambda/\Lambda_0)]$.
- Contributions of high-energy sector can be taken into account by *reparameterisation* of original theory. One needs to assume that the (unobservable) bare mass / charge are infinite to get finite results.
- A theory is renormalisable if this is possible for all orders, which is not true for many quantum field theories!

Realistic Interpretation: The cut-Off parameter Λ represents upper bound of energy scale to which the EFT applies: For higher energies, the theory is not applicable.

It is possible to get to a theory with a higher cut-off by solving the renormalisation group equations.

Decoupling theorem (simple case):

- For two coupled systems with energy scales $m_2 > m_1$ and described by a renormalisable theory, there is a renormalisation condition according to which the effects of physics at scale m_2 can be effectively included in theory with scale m_1 by changing parameters.
- Implies existence of EFT at m_1 which is not applicable to m_2 .
- Legitimacy of non-renormalisable theories: Even if theory of m_1 is not renormalisable, the physics of m_2 can be included.
- Assumptions are not always fulfilled! For instance, there is not always a renormalisable theory of the composite system, and mass scales might not always separate neatly.

1.3 Bottom-up / top-down approach

Bottom-up approach:

- Either there is no theory, or there is a theory for Λ_1 , and we try to find a theory for $\Lambda_2 > \Lambda_1$.
- No new particles: Theories are empirically equivalent, but differ in values for mass and charge parameters.
- New particles: Mass and charge of original particles adapted by renormalisation equations, new particles have to be identified (mass, charge, fermions/bosons, coupling).
 $\Rightarrow$ Theoretical and experimental considerations are necessary.
- Example: Fermi theory.

Top-down approach:

- We Start with a more fundamental theory valid on Λ_1 , and try to construct a theory for $\Lambda_2 < \Lambda_1$.
- There is a systematic procedure if the original theory is renormalisable and the decoupling theorem holds.
- Example: Case Study (next section).

2 Case Study

2.1 QCD

The Lagrangian for quantum chromodynamics (QCD):

$$\mathcal{L}_{QCD} = \bar{\psi} (i\gamma_\mu D^\mu - \hat{m}_0) \psi - \frac{1}{4} F_{k\mu\nu} F_k^{\mu\nu}. \quad (4)$$

Exact consequences can be derived by lattice gauge theory (for low energy): Quark and gluon fields are defined on a fictional lattice with finite spacing $a > 0$, then we consider $a \rightarrow 0$. This method yields accurate results, but does not provide an understanding of the described phenomena.

To obtain this understanding, models and EFTs are used. They reveal the following features of QCD:

1. QCD is asymptotically free (quarks move freely at very high energies).
2. QCD exhibits quark confinement (there are several accounts of what exactly this amounts to).

3. Low-energy QCD is (almost) chirally invariant and exhibits dynamical chiral symmetry breaking: Chiral partners do not seem to exist (exp. proton).

2.2 Models

Model: set of assumptions (+ diagrams, ...) that might be inspired by a theory or specify the concrete object/system. Often used to isolate assumptions of the theory. We explore the consequences of these assumptions. Models are incredibly useful, but the relation to theories is not fully understood.

Example: Bag models explore consequences of quark confinement. Bohr model ignores the fact that moving charges would emit radiation.

2.3 EFTs

Example Chiral Perturbation Theory: We expand the Lagrangian of QCD for momentum in the discussed process. We replace quark and gluon fields of QCD by Goldstone bosons represented as 2×2 matrices $U(x) \in SU(2)$ (this is possible because of dynamical breaking of chiral symmetry). Then:

$$\mathcal{L}_{QCD} = \mathcal{L}_{eff}(U, \partial U, \partial^2 U, \dots) . \quad (5)$$

Power-series expansion in momentum and mass leads to:

$$\mathcal{L}_{eff}(U, \partial U) . \quad (6)$$

This effective Lagrangian is much easier to treat, but still gives accurate results.

2.4 Theories, Models, EFTs

Functions:

1. Theories: Wide scope of applicability, do not provide local understanding of relevant physical processes.
2. Models: Produce local understanding, easier to handle mathematically, assumptions lack deeper foundation.
3. EFTs: More adjustable parameters than a model of the same system, work well only if energy scales separate well, close relation to framework of quantum field theory.

Relations:

1. Theories and models are often not related deductively in any direct sense, theories can inspire models, models often isolate some of theories' assumptions.
2. Models and EFTs are not always easy to distinguish, some EFTs are treated like models.
3. Theories and EFTs are sometimes related in a deductive sense.

3 Philosophical Implications

3.1 Pluralism and coherence

A final fundamental theory would have three problems:

- It is unlikely to provide local understanding of observed phenomena.
- It cannot be applied without various assumptions about the system.
- It is impossible to arrive at a final theory without extensive use of models and EFTs.

⇒ EFTs and models occupy an important place in scientific research towards and even given a final theory!

Example: Quantum Mechanics does not directly provide local understanding of many phenomena. Various assumptions must be made about the system (atoms, molecules, crystal lattices) to make predictions. Quantum mechanics was developed by use of models and effective theories.

⇒ We should expect the same for a final theory.

Do either models or EFTs instead make the other two obsolete, or should “dominate” scientific inquiry?

- Models need to be connected to theories and theories are needed to inspire models.
- If EFTs are constructed from theories, those theories are still necessary. If not, a theory can still provide valuable insight that a set of EFTs does not.

⇒ All three tools are necessary!

Their exact relation is unclear: They are often logically inconsistent in a straightforward sense, but there are still some deductive, ontological or semantic relations.

Could this relation between a model M and a theory T be modelled by conditional probability $P(M|T)$? How does this work if M and T are contradictory (for instance: Bohr model and classical electrodynamics)?

3.2 Reduction and emergence

Ontological pluralism: Empirical reality “seems to be organised in a multitude [...] or quasi-autonomous layers. Each layer has its own ontology, and this ontology is to a considerable

extent independent of the physics at higher energies.”

⇒ This should be a pragmatic thesis at best! EFTs might instead be an interesting case of emergence. Furthermore, ontological pluralism in this sense seems to be based on the decoupling theorem, which only holds if there *is* already an underlying renormalisable theory of the system. This might be in conflict with ontological pluralism.

Epistemological antifoundationalism:

Layers have their own ontology, but also their own fundamental theory. There is no fundamental theory for all layers, and there can never be a final theory.

⇒ Not clear why this should be the case! The fact that a final theory does not seem *near* is not enough reason to reject its possibility.

Methodological antireductionism: Favors bottom-up strategy. This might be useful. However, it is again not clear that we should reject the possibility of a final theory. As discussed before, such a theory does not devalue EFTs or models before and after it.

3.3 Explanation and understanding

Scientific explanation should show:

1. How the phenomenon reached the state considered.
2. The place of the phenomenon in a larger framework.

It is not clear if both can be achieved by the same theory / model / EFT.

Different accounts focus on different components:

- Causal/mechanical account: Providing mechanism which produces the effect under consideration (shows 1, lends itself well to models and EFTs). $\Rightarrow$ Local understanding.
- Unification: Explanation places phenomenon in general framework (shows 2, lends itself well to theories). $\Rightarrow$ Global understanding.

$\Rightarrow$ Both are legitimate as scientific explanations! Explanations can be given from different perspectives (EFTs, models, theories, 1, 2). The best explained phenomena have explanations from multiple or all of these.

4 Questions and Takeaways

1. Theories, models and EFTs are clearly all incredibly useful and indispensable tools for scientific investigation. We should not discount their importance by overly restrictive notions of explanation or understanding, or by prematurely rejecting the possibility of a final theory.
2. What exactly is the difference between local / global understanding? Is it true that theories usually provide local and models global understanding?
3. The question still remains how we should interpret the content of EFTs and whether we think they commit us to any ontological claims and how this relates to the underlying theory, if there is any. Do EFTs lend themselves to an instrumentalist interpretation?
4. What conception of reality and layers of reality is supported by EFTs?
5. What can EFTs tell us about emergence?
6. What is the status of axiomatic QFT?